right side of the piston is sent back to the reservoir through A and E.

Quick return of the ram

Due to the change of stroke volume of the cylinder during forward and return stroke the piston returns at a faster rate. The stroke volume of the cylinder during return stroke is comparatively smaller than during forward stroke due to the presence of the piston rod. So the pressure increases automatically to push the piston and the tool head at a faster speed during return stroke.

8.16 Advantages of Hydraulic Drive over Mechanical Drive

1. Producing of heat is removed, when mechanical force is applied
2. It has high efficiency. No sound is produced
3. Always pressure is even
4. Hydraulic equipments are stable
5. No need of locking the hydraulic equipments
6. Withstand the overload accurately
7. Piston is functioned without decreasing of pressure inside, even if the flow of liquid is stopped in hydraulic equipment
8. Hydraulic energy can be transformed easily to desired place
9. It is easy to operate the control equipments of hydraulic drive
10. There is no need of belt, chain, clutch, brake, cam, lever in hydraulic drive
11. It is easy to change the speed of rotational, horizontal, continuous, or constant force of work in hydraulic drive
12. It is reliable and acts rapidly
13. It is very simple to form
14. Low space is enough
15. It can be done the work at desired place even if the hydraulic equipment is in long distance
16. It is possible to change the direction of rotational or linear movement of hydraulic equipment in working condition
17. Rotational or linear speed may be changed or not, according to our head
18. It is possible to get minimum force with minimum energy
19. Easy to design machines, when using of hydraulic equipment
20. Erection and maintenance is easy
21. If using of electrical equipments caused danger, we may use hydraulic equipments instead of electrical equipments.

Disadvantages

- Leakage of oil causes dust and dirt
- Hydraulic pipes may be damaged due to dust
- If the pressure of flow oil exceeds the limit, hydraulic pipe may be burst.
- Due to leakage of oil, there may be causing of fire accident

ACTIVITIES

1. To arrange the students to know about the working principle of hydraulic equipments by visiting polytechnic or Engineering college.

XII BME_English version CHAPTER 08.indd 153 1/8/2020 1:39:35 PM